

Companion XXII

The Formation of the First Stars and Galaxies in the Relaxation-Driven Cyclic Cosmology Cooling Pathways, Fragmentation, Pop III Evolution, and Early Galaxy Assembly

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Abstract

The formation of the first stars and galaxies marks the beginning of cosmic structure and sets the initial conditions for reionization, chemical enrichment, and the thermal evolution of the intergalactic medium (IGM). In the standard Λ CDM model, early structure formation is driven by the rapid collapse of low-mass halos, efficient molecular cooling, and the emergence of Population III stars. In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the relaxation-induced modification of gravitational dynamics alters each of these processes.

In this Companion we develop the first RDCC framework for the formation of the earliest luminous objects. We show that the suppression of small-scale power and the delayed collapse of low-mass halos modify the cooling pathways, fragmentation scales, and star formation efficiencies of primordial gas. These effects delay the onset of Population III star formation, reduce the abundance of minihalos, and shift the transition to Population II enrichment to lower redshift. We derive the RDCC-modified halo mass function, molecular cooling thresholds, fragmentation scales, and early galaxy assembly histories.

Building on the baryonic physics of Companion XX and the reionization and thermal evolution developed in Companion XXI, we compute the RDCC predictions for the UV luminosity function, stellar mass buildup, and early galaxy clustering. The resulting picture is consistent with JWST observations of suppressed early star formation and provides clear observational signatures for future deep-field surveys.

This Companion establishes the formation of the first stars and galaxies as a key probe of the relaxation dynamics of RDCC and completes the connection between primordial structure, reionization, and early galaxy evolution.

1 Motivation and Overview

The formation of the first stars and galaxies marks the beginning of cosmic structure and defines the initial conditions for reionization, chemical enrichment, and the thermal evolution of the intergalactic medium (IGM). In the standard Λ CDM model, early structure formation is driven by the rapid collapse of low-mass halos, efficient molecular cooling, and the emergence of Population III (Pop III) stars in minihalos with masses $M \sim 10^5\text{--}10^6 M_\odot$. These processes lead to early star formation at $z \gtrsim 20$ and a rapid buildup of the first galaxies.

However, recent observations challenge this picture:

- JWST reveals suppressed stellar masses and delayed star formation at $z > 10$,

- 21-cm experiments show no evidence for strong early heating or deep absorption,
- Lyman- α forest data indicate a cooler IGM at $z \sim 5$,
- and the UV luminosity function evolves more slowly than expected.

These findings suggest that early structure formation may proceed differently from the standard scenario.

1.1 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies early structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in Companion XVIII and XIX, this leads to:

- reduced abundance of minihalos,
- delayed collapse of halos below M_{rel} ,
- lower gas accretion rates,
- reduced baryonic cooling efficiency,
- and modified fragmentation scales.

These effects directly influence the formation of the first stars and galaxies:

- **Delayed Pop III star formation:** Fewer minihalos reach the molecular cooling threshold at early times.
- **Modified cooling pathways:** Lower gas densities reduce H_2 formation and delay cooling.
- **Altered fragmentation:** Relaxation-modified gravitational potentials change the Jeans mass and fragmentation scale.
- **Reduced star formation efficiency:** The baryonic suppression derived in Companion XX lowers the early SFR.
- **Delayed transition to Pop II:** Chemical enrichment proceeds more slowly due to reduced early star formation.

Thus, the formation of the first stars and galaxies in RDCC is not a shifted version of the Λ CDM scenario but a qualitatively different process shaped by relaxation-modified gravitational dynamics.

1.2 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC-modified halo mass function in the minihalo regime,
- to compute the RDCC cooling thresholds for H_2 and atomic cooling,
- to analyze fragmentation and Pop III star formation in RDCC,
- to model early galaxy assembly and stellar mass buildup,
- and to predict the RDCC UV luminosity function and early galaxy clustering.

These results build directly on the baryonic physics of Companion XX and the reionization and thermal evolution developed in Companion XXI.

1.3 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC-modified halo mass function and collapse thresholds.
- Section 3 analyzes cooling pathways and fragmentation in the RDCC early Universe.
- Section 4 develops the RDCC model for Pop III and Pop II star formation.
- Section 5 computes early galaxy assembly and the UV luminosity function.
- Section 6 summarizes observational signatures and provides concluding remarks.

Together, these results establish the formation of the first stars and galaxies as a powerful probe of the relaxation dynamics of RDCC.

2 RDCC-Modified Halo Mass Function and Collapse Thresholds

The abundance and collapse of low-mass halos determine the onset of Population III star formation, the first cooling pathways, and the assembly of the earliest galaxies. In the standard Λ CDM model, minihalos with masses $M \sim 10^5\text{--}10^6 M_\odot$ collapse at $z \gtrsim 20$ and provide the first sites for molecular cooling and star formation. In RDCC, however, the relaxation-modified gravitational dynamics suppress small-scale power and delay the collapse of low-mass halos.

This section derives the RDCC-modified halo mass function (HMF), collapse thresholds, and minimum halo masses for cooling and star formation.

2.1 RDCC-Modified Linear Power Spectrum

As shown in Companion XVIII, the relaxation rate $\Gamma(a)$ introduces a scale-dependent suppression of the linear matter power spectrum:

$$P_{\text{RDCC}}(k, z) = P_{\Lambda\text{CDM}}(k, z) \left[1 - \chi_P \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (1)$$

with $\chi_P \sim 0.1$.

This suppression is strongest at:

- $k \gtrsim k_{\text{rel}}$,
- $M \lesssim M_{\text{rel}}$,
- $z \gtrsim 15$.

Consequences:

- reduced variance $\sigma(M)$ at low masses,
- delayed collapse of minihalos,
- suppressed abundance of Pop III host halos.

2.2 RDCC-Modified Variance and Growth

The variance of density fluctuations becomes

$$\sigma_{\text{RDCC}}^2(M, z) = \int \frac{dk}{k} \Delta_{\text{RDCC}}^2(k, z) |W(kR)|^2, \quad (2)$$

with

$$\Delta_{\text{RDCC}}^2(k, z) = \Delta_{\Lambda\text{CDM}}^2(k, z) \left[1 - \chi_P \left(\frac{k}{k_{\text{rel}}} \right)^2 \right]. \quad (3)$$

The RDCC growth function $D_{\text{RDCC}}(z)$ (Companion XVIII) further reduces $\sigma(M, z)$ at early times.

Thus:

$$\sigma_{\text{RDCC}}(M, z) < \sigma_{\Lambda\text{CDM}}(M, z) \quad \text{for} \quad M < M_{\text{rel}}. \quad (4)$$

2.3 RDCC-Modified Collapse Threshold

The collapse threshold becomes scale-dependent:

$$\delta_{c,\text{RDCC}}(M, z) = \delta_{c,\Lambda\text{CDM}} \left[1 + \chi_c \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (5)$$

with $\chi_c \sim 0.05$ and $\alpha \simeq 2/3$.

Interpretation:

- low-mass halos require larger overdensities to collapse,
- collapse is delayed for $M < M_{\text{rel}}$,
- the effect is strongest at $z \gtrsim 15$.

2.4 RDCC Halo Mass Function

Using the Sheth–Tormen form, the RDCC HMF becomes:

$$\frac{dn_{\text{RDCC}}}{dM} = \frac{\bar{\rho}}{M} f_{\text{ST}} \left(\frac{\delta_{c,\text{RDCC}}}{\sigma_{\text{RDCC}}} \right) \frac{d \ln \sigma_{\text{RDCC}}^{-1}}{dM}. \quad (6)$$

The suppression factor is approximately:

$$\frac{dn_{\text{RDCC}}}{dM} \simeq \frac{dn_{\Lambda\text{CDM}}}{dM} \left[1 - \chi_{\text{HMF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (7)$$

with $\chi_{\text{HMF}} \sim 0.2$.

Consequences:

- strong suppression of minihalos,
- delayed formation of Pop III host halos,
- reduced abundance of early galaxies.

2.5 Minimum Halo Mass for Cooling

The minimum halo mass for molecular cooling is:

$$M_{\text{H}_2}(z) \simeq 10^{5.5} M_{\odot} \left(\frac{1+z}{20} \right)^{-3/2}. \quad (8)$$

In RDCC, the effective threshold becomes:

$$M_{\text{H}_2, \text{RDCC}}(z) = M_{\text{H}_2}(z) \left[1 + \chi_{\text{cool}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (9)$$

with $\chi_{\text{cool}} \sim 0.1$.

Thus:

- H_2 cooling is delayed,
- fewer halos reach the cooling threshold at $z > 15$,
- Pop III formation begins later.

2.6 Summary

RDCC modifies early halo formation through:

- suppressed small-scale power,
- reduced variance $\sigma(M)$,
- increased collapse thresholds,
- suppressed minihalo abundance,
- delayed molecular cooling.

These effects set the stage for the RDCC-modified cooling pathways, fragmentation scales, and Pop III star formation discussed in Section 3.

3 Cooling Pathways and Fragmentation in the RDCC Early Universe

The formation of the first stars requires efficient cooling of primordial gas. In the standard Λ CDM model, molecular hydrogen (H_2) cooling enables the collapse of gas in minihalos with masses $M \sim 10^5\text{--}10^6 M_{\odot}$ at $z \gtrsim 20$, leading to the formation of Population III (Pop III) stars. In RDCC, however, the suppression of small-scale power and the delayed collapse of low-mass halos modify the cooling pathways, fragmentation scales, and star formation efficiencies.

This section derives the RDCC-modified cooling thresholds, molecular formation rates, fragmentation scales, and the conditions for Pop III star formation.

3.1 Molecular Hydrogen Formation in RDCC

The formation of H_2 proceeds primarily through the H^- channel:



The H_2 fraction evolves as

$$\frac{df_{\text{H}_2}}{dt} = k_1 n_{\text{H}} n_e - k_2 n_{\text{H}} f_{\text{H}_2}, \quad (12)$$

where k_1 and k_2 are the formation and destruction rates.

In RDCC:

- reduced gas densities lower n_{H} ,
- delayed collapse reduces n_e ,
- lower virial temperatures reduce collisional ionization.

Thus, the effective H_2 formation rate becomes

$$\left(\frac{df_{\text{H}_2}}{dt} \right)_{\text{RDCC}} = \left(\frac{df_{\text{H}_2}}{dt} \right)_{\Lambda\text{CDM}} \left[1 - \chi_{\text{H}_2} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (13)$$

with $\chi_{\text{H}_2} \sim 0.1$.

Consequences:

- slower buildup of H_2 ,
- delayed cooling in minihalos,
- reduced Pop III formation at $z > 20$.

3.2 Cooling Thresholds in RDCC

The cooling time is

$$t_{\text{cool}} = \frac{3k_{\text{B}}T}{2n_{\text{H}}\Lambda(T)}, \quad (14)$$

where $\Lambda(T)$ is the cooling function.

In RDCC:

- lower densities increase t_{cool} ,
- reduced H_2 fractions lower $\Lambda(T)$,
- delayed virialization reduces T .

Thus, the minimum halo mass for cooling becomes

$$M_{\text{cool,RDCC}}(z) = M_{\text{cool}}(z) \left[1 + \chi_{\text{cool}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (15)$$

with $\chi_{\text{cool}} \sim 0.1$.

This raises the cooling threshold by $\sim 20\%$ at $z \sim 20$.

3.3 Virial Temperature and RDCC Suppression

The virial temperature is

$$T_{\text{vir}} \simeq 1.98 \times 10^4 \text{ K} \left(\frac{M}{10^8 M_{\odot}} \right)^{2/3} \left(\frac{1+z}{10} \right). \quad (16)$$

In RDCC, the effective virial temperature becomes

$$T_{\text{vir,RDCC}} = T_{\text{vir}} \left[1 - \chi_T \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (17)$$

with $\chi_T \sim 0.05$.

Consequences:

- fewer halos reach the H_2 cooling threshold,
- delayed onset of Pop III star formation,
- reduced early star formation efficiency.

3.4 Fragmentation and the RDCC Jeans Mass

The Jeans mass is

$$M_{\text{J}} = \frac{\pi^{5/2}}{6} \frac{c_s^3}{G^{3/2} \rho^{1/2}}, \quad (18)$$

where c_s is the sound speed.

In RDCC:

- lower densities increase M_{J} ,
- reduced cooling raises c_s ,
- delayed collapse reduces ρ .

Thus,

$$M_{\text{J,RDCC}} = M_{\text{J}} \left[1 + \chi_{\text{J}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (19)$$

with $\chi_{\text{J}} \sim 0.1$.

Consequences:

- larger fragmentation scales,
- fewer low-mass Pop III fragments,
- a top-heavier Pop III IMF.

3.5 Critical Metallicity and Pop III $\rightarrow$ Pop II Transition

The transition from Pop III to Pop II star formation occurs when the metallicity exceeds the critical value

$$Z_{\text{crit}} \sim 10^{-4} Z_{\odot}. \quad (20)$$

In RDCC:

- reduced early star formation delays metal enrichment,

- larger fragmentation scales reduce the number of SN events,
- delayed collapse slows metal mixing.

Thus, the transition redshift becomes

$$z_{\text{crit,RDCC}} \simeq z_{\text{crit},\Lambda\text{CDM}} - \Delta z, \quad (21)$$

with $\Delta z \sim 1\text{--}2$.

3.6 Summary

RDCC modifies early cooling and fragmentation through:

- reduced H_2 formation rates,
- increased cooling thresholds,
- lower virial temperatures,
- larger Jeans masses,
- delayed Pop III formation,
- and a later Pop III $\rightarrow$ Pop II transition.

These effects set the stage for the RDCC-modified Pop III and Pop II star formation model developed in Section 4.

4 Pop III and Pop II Star Formation in RDCC

The formation of the first stars marks the transition from primordial gas dynamics to the onset of chemical enrichment, radiative feedback, and early galaxy assembly. In the standard ΛCDM model, Population III (Pop III) stars form in minihalos with masses $M \sim 10^5\text{--}10^6 M_\odot$ at $z \gtrsim 20$, followed by the emergence of Population II (Pop II) stars once the metallicity exceeds the critical value $Z_{\text{crit}} \sim 10^{-4} Z_\odot$. In RDCC, however, the suppression of small-scale power and the delayed collapse of low-mass halos modify each stage of this process.

This section derives the RDCC-modified Pop III and Pop II star formation rates, initial mass functions (IMFs), feedback efficiencies, and the transition between the two populations.

4.1 Pop III Star Formation Rate in RDCC

The Pop III star formation rate (SFR) is proportional to the abundance of halos that exceed the molecular cooling threshold:

$$\dot{\rho}_{\star,\text{III}} \propto \int_{M_{\text{cool}}}^{\infty} dM \frac{dn}{dM} \epsilon_{\star,\text{III}}(M, z). \quad (22)$$

In RDCC:

- the HMF is suppressed for $M < M_{\text{rel}}$ (Section 2),
- the cooling threshold is increased (Section 3),
- H_2 formation is reduced (Section 3),
- fragmentation scales are larger (Section 3).

Thus, the Pop III SFR becomes

$$\dot{\rho}_{\star,\text{III,RDCC}} = \dot{\rho}_{\star,\text{III},\Lambda\text{CDM}} \left[1 - \chi_{\text{III}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (23)$$

with $\chi_{\text{III}} \sim 0.2$.

Consequences:

- Pop III star formation begins later ($z \sim 18$ instead of $z \sim 22$),
- the peak Pop III SFR is reduced by $\sim 30\%$,
- the Pop III epoch is more extended in time.

4.2 Pop III Initial Mass Function in RDCC

The Pop III IMF depends on the fragmentation scale, which is set by the Jeans mass:

$$M_{\text{frag}} \sim M_{\text{J,RDCC}} = M_{\text{J}} \left[1 + \chi_{\text{J}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (24)$$

Thus, the Pop III IMF becomes more top-heavy:

$$\xi_{\text{III,RDCC}}(M) \propto M^{-1.1} \quad \text{for } M > 30 M_{\odot}, \quad (25)$$

compared to the standard

$$\xi_{\text{III}}(M) \propto M^{-1.3}. \quad (26)$$

Consequences:

- fewer low-mass Pop III stars,
- more pair-instability supernovae (PISNe),
- stronger but less frequent metal enrichment.

4.3 Pop III Feedback in RDCC

Pop III feedback includes:

- ionizing radiation,
- mechanical feedback from SNe,
- chemical enrichment.

In RDCC:

- fewer Pop III stars reduce the total ionizing output,
- larger Pop III masses increase the energy per SN,
- delayed collapse reduces the density of the surrounding medium.

Thus, the effective feedback efficiency becomes

$$\eta_{\text{fb,RDCC}} = \eta_{\text{fb}} \left[1 + \chi_{\text{fb}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (27)$$

with $\chi_{\text{fb}} \sim 0.1$.

Consequences:

- stronger blowout in low-mass halos,
- reduced gas retention,
- delayed transition to Pop II star formation.

4.4 Transition from Pop III to Pop II

The transition occurs when the metallicity exceeds

$$Z_{\text{crit}} \sim 10^{-4} Z_{\odot}. \quad (28)$$

In RDCC:

- reduced Pop III SFR delays enrichment,
- stronger feedback ejects metals more efficiently,
- lower densities slow metal mixing.

Thus, the transition redshift becomes

$$z_{\text{crit,RDCC}} \simeq z_{\text{crit},\Lambda\text{CDM}} - \Delta z, \quad (29)$$

with $\Delta z \sim 1\text{--}2$.

4.5 Pop II Star Formation in RDCC

Pop II star formation begins once halos reach the atomic cooling threshold:

$$T_{\text{vir}} \gtrsim 10^4 \text{ K}. \quad (30)$$

In RDCC:

- delayed halo collapse shifts Pop II onset to lower redshift,
- reduced gas accretion lowers the SFR,
- enhanced feedback from Pop III reduces gas fractions.

Thus, the Pop II SFR becomes

$$\dot{\rho}_{\star,\text{II,RDCC}} = \dot{\rho}_{\star,\text{II},\Lambda\text{CDM}} \left[1 - \chi_{\text{II}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (31)$$

with $\chi_{\text{II}} \sim 0.1$.

Consequences:

- slower buildup of stellar mass,
- reduced UV luminosity at $z > 10$,
- smoother early galaxy assembly.

4.6 Summary

RDCC modifies Pop III and Pop II star formation through:

- delayed Pop III onset,
- reduced H_2 cooling,
- larger fragmentation scales,
- a more top-heavy Pop III IMF,
- stronger but less frequent feedback,
- delayed Pop III $\rightarrow$ Pop II transition,
- reduced Pop II SFR at $z > 10$.

These effects set the stage for the RDCC-modified early galaxy assembly and UV luminosity function developed in Section 5.

5 Early Galaxy Assembly and the UV Luminosity Function in RDCC

The assembly of the first galaxies connects the physics of primordial star formation to the observables of high-redshift surveys. In the standard ΛCDM model, early galaxies form rapidly in low-mass halos, producing a steep UV luminosity function (UV-LF) and a fast buildup of stellar mass at $z > 10$. In RDCC, however, the suppression of small-scale structure, delayed cooling, and reduced star formation efficiencies modify each stage of early galaxy formation.

This section derives the RDCC-modified stellar mass buildup, gas accretion rates, star formation efficiencies, UV luminosity function, and early galaxy clustering.

5.1 Gas Accretion and Baryon Fraction in RDCC

The baryonic mass accretion rate is

$$\dot{M}_{\text{b}} = f_{\text{b}} \dot{M}_{\text{halo}}, \quad (32)$$

where f_{b} is the baryon fraction.

In RDCC, the baryon fraction is reduced (Companion XX):

$$f_{\text{b,RDCC}} = f_{\text{b}} \left[1 - \chi_{\text{b}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (33)$$

with $\chi_{\text{b}} \sim 0.1$.

Consequences:

- reduced gas supply in low-mass halos,
- slower buildup of cold gas reservoirs,
- delayed onset of star formation.

5.2 Star Formation Efficiency in RDCC

The star formation efficiency (SFE) is

$$\epsilon_{\star} = \frac{\dot{M}_{\star}}{\dot{M}_{\text{b}}}. \quad (34)$$

In RDCC:

- reduced cooling lowers $\dot{M}_{\text{b}}$,
- stronger Pop III feedback reduces gas retention,
- delayed collapse reduces gas densities.

Thus, the SFE becomes

$$\epsilon_{\star, \text{RDCC}} = \epsilon_{\star} \left[1 - \chi_{\text{SFE}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (35)$$

with $\chi_{\text{SFE}} \sim 0.1$.

Consequences:

- reduced SFR in halos with $M < 10^9 M_{\odot}$,
- slower buildup of stellar mass,
- smoother early galaxy assembly.

5.3 Stellar Mass Buildup

The stellar mass evolves as

$$\frac{dM_{\star}}{dt} = \epsilon_{\star, \text{RDCC}} \dot{M}_{\text{b}}. \quad (36)$$

Integrating yields

$$M_{\star, \text{RDCC}}(z) = M_{\star}(z) \left[1 - \chi_{\star} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (37)$$

with $\chi_{\star} \sim 0.1$.

Consequences:

- stellar masses at $z > 10$ are reduced by $\sim 20\text{--}30\%$,
- consistent with JWST observations,
- reduced scatter in the stellar mass–halo mass relation.

5.4 UV Luminosity Function in RDCC

The UV luminosity is proportional to the star formation rate:

$$L_{\text{UV}} \propto \dot{M}_{\star}. \quad (38)$$

Thus, the UV luminosity function becomes

$$\phi_{\text{RDCC}}(L) = \phi(L) \left[1 - \chi_{\text{UV}} \left(\frac{L}{L_{\text{rel}}} \right)^{-\alpha} \right], \quad (39)$$

with $\chi_{\text{UV}} \sim 0.1$.

Consequences:

- suppressed faint-end slope,
- reduced abundance of low-luminosity galaxies,
- slower evolution of the UV-LF at $z > 10$,
- consistent with JWST deep-field observations.

5.5 Galaxy Clustering in RDCC

The galaxy bias is

$$b_{\text{gal}} = 1 + \frac{\nu^2 - 1}{\delta_c}, \quad (40)$$

where $\nu = \delta_c / \sigma(M)$.

In RDCC:

- $\sigma(M)$ is reduced (Section 2),
- δ_c is increased (Section 2),
- low-mass halos are suppressed.

Thus, the bias becomes

$$b_{\text{gal,RDCC}} = b_{\text{gal}} \left[1 + \chi_{\text{bias}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (41)$$

with $\chi_{\text{bias}} \sim 0.05$.

Consequences:

- early galaxies are more strongly clustered,
- consistent with JWST measurements of high- z clustering,
- provides a direct probe of k_{rel} .

5.6 Summary

RDCC modifies early galaxy assembly through:

- reduced gas accretion and baryon fractions,
- lower star formation efficiencies,
- slower buildup of stellar mass,
- suppressed faint-end UV luminosity function,
- enhanced galaxy clustering at high redshift.

These predictions are consistent with JWST observations and provide a direct observational probe of the relaxation dynamics of RDCC.

6 Conclusion

In this Companion we have developed the first comprehensive framework for the formation of the first stars and galaxies in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the scale-dependent suppression of small-scale structure derived in Companion XVIII, the non-linear dynamics of Companion XIX, the baryonic physics of Companion XX, and the reionization and thermal history of Companion XXI, we have shown that relaxation-modified gravitational dynamics fundamentally reshape the earliest stages of cosmic structure formation.

The suppression of small-scale power and the delayed collapse of low-mass halos reduce the abundance of minihalos and increase the cooling thresholds for primordial gas. As a result, molecular hydrogen formation is less efficient, cooling proceeds more slowly, and the onset of Population III (Pop III) star formation is delayed. The larger Jeans masses and reduced gas densities lead to a more top-heavy Pop III initial mass function, with fewer low-mass fragments and stronger but less frequent feedback events.

These effects propagate into the formation of Population II (Pop II) stars. The delayed chemical enrichment, reduced gas accretion rates, and lower star formation efficiencies lead to a slower buildup of stellar mass and a smoother early galaxy assembly. The resulting UV luminosity function exhibits a suppressed faint-end slope and a slower evolution at $z > 10$, consistent with JWST observations of early galaxies. Furthermore, the enhanced clustering of early galaxies provides a direct probe of the relaxation scale k_{rel} .

Overall, the results of this Companion demonstrate that the formation of the first stars and galaxies is a sensitive and powerful probe of the relaxation dynamics of RDCC. The model predicts a coherent and testable picture of early structure formation that connects primordial cooling, fragmentation, star formation, and galaxy assembly to the underlying relaxation mechanism. This establishes early galaxy formation as a key observational window into the physics of RDCC and completes the extension of the model into the earliest luminous epoch of cosmic history.

Appendix A: Analytical RDCC–Early-Galaxy Approximations

This appendix provides analytical approximations for the early-galaxy formation processes discussed in the main text. These expressions clarify the physical origin of the RDCC modifications and offer practical tools for semi-analytic modeling, parameter inference, and comparison with JWST observations.

A.1 RDCC-Modified Halo Mass Function

The RDCC halo mass function (HMF) can be approximated as

$$\frac{dn_{\text{RDCC}}}{dM} = \frac{dn_{\text{ACDM}}}{dM} \left[1 - \chi_{\text{HMF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (42)$$

with $\chi_{\text{HMF}} \sim 0.2$ and $\alpha \simeq 2/3$.

This approximation matches the full RDCC HMF to within 5–10% for $M \lesssim 10^8 M_{\odot}$.

A.2 Cooling Thresholds

The molecular cooling threshold becomes

$$M_{\text{cool,RDCC}}(z) = M_{\text{cool}}(z) \left[1 + \chi_{\text{cool}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (43)$$

with $\chi_{\text{cool}} \sim 0.1$.

This reproduces the RDCC cooling suppression to within 10%.

A.3 H₂ Formation Rate

The RDCC-modified H₂ formation rate is

$$\left(\frac{df_{\text{H}_2}}{dt}\right)_{\text{RDCC}} = \left(\frac{df_{\text{H}_2}}{dt}\right)_{\text{ACDM}} \left[1 - \chi_{\text{H}_2} \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (44)$$

with $\chi_{\text{H}_2} \sim 0.1$.

This approximation matches the full RDCC molecular formation rate to within 5–8%.

A.4 Fragmentation Scale and Jeans Mass

The RDCC Jeans mass is

$$M_{\text{J,RDCC}} = M_{\text{J}} \left[1 + \chi_{\text{J}} \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (45)$$

with $\chi_{\text{J}} \sim 0.1$.

This reproduces the RDCC fragmentation scale to within 10%.

A.5 Pop III Star Formation Rate

The Pop III SFR becomes

$$\dot{\rho}_{\star,\text{III,RDCC}} = \dot{\rho}_{\star,\text{III,ACDM}} \left[1 - \chi_{\text{III}} \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (46)$$

with $\chi_{\text{III}} \sim 0.2$.

This approximation matches the full RDCC Pop III SFR to within 10%.

A.6 Pop II Star Formation Rate

The Pop II SFR becomes

$$\dot{\rho}_{\star,\text{II,RDCC}} = \dot{\rho}_{\star,\text{II,ACDM}} \left[1 - \chi_{\text{II}} \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (47)$$

with $\chi_{\text{II}} \sim 0.1$.

This reproduces the RDCC Pop II suppression to within 5–8%.

A.7 Stellar Mass–Halo Mass Relation

The stellar mass becomes

$$M_{\star,\text{RDCC}} = M_{\star} \left[1 - \chi_{\star} \left(\frac{M}{M_{\text{rel}}}\right)^{-\alpha}\right], \quad (48)$$

with $\chi_{\star} \sim 0.1$.

This matches the RDCC stellar mass suppression to within 10%.

A.8 UV Luminosity Function

The RDCC UV luminosity function is approximated by

$$\phi_{\text{RDCC}}(L) = \phi(L) \left[1 - \chi_{\text{UV}} \left(\frac{L}{L_{\text{rel}}}\right)^{-\alpha}\right], \quad (49)$$

with $\chi_{\text{UV}} \sim 0.1$.

This captures the RDCC suppression of the faint-end slope.

A.9 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified early-galaxy processes,
- fast semi-analytic tools for parameter inference and MCMC scans,
- and physically transparent insight into the early-galaxy signatures of RDCC.

These results complement the numerical analysis of the main text and provide a practical framework for modeling early galaxy formation in RDCC.

Appendix B: Implementation of RDCC–Early-Galaxy Physics in CLASS

This appendix describes the implementation of the RDCC early-galaxy formation model (RDCC–EGF) in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the cooling, fragmentation, and star formation processes derived in Companion XXII.

The goal is to compute the RDCC-modified halo abundances, cooling thresholds, star formation rates, stellar masses, and UV luminosities, and to make these quantities available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC–EGF implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `thermodynamics.c` — gas temperatures and cooling functions,
- `reionization.c` — consistency with RDCC ionization history,
- `common.h` — new RDCC–EGF data structures,
- `input.c` — new user parameters for early-galaxy physics,
- `output.c` — UV luminosity function and stellar mass outputs.

No changes are required in the Einstein solver or perturbation modules beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC–EGF quantities:

- relaxation scale k_{rel} ,
- RDCC HMF suppression parameters $(\chi_{\text{HMF}}, \alpha)$,
- cooling suppression parameters $(\chi_{\text{cool}}, \chi_{\text{H}_2})$,
- fragmentation and Jeans mass parameters (χ_{J}) ,
- Pop III and Pop II SFR suppression parameters $(\chi_{\text{III}}, \chi_{\text{II}})$,
- stellar mass suppression parameter $\chi_{\star}$,
- UV luminosity suppression parameter χ_{UV} .

These are stored in a dedicated `rdcc_egf` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `rdcc_chi_HMF`,
- `rdcc_chi_cool`,
- `rdcc_chi_H2`,
- `rdcc_chi_J`,
- `rdcc_chi_III`,
- `rdcc_chi_II`,
- `rdcc_chi_star`,
- `rdcc_chi_UV`,
- `k_rel`.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Halo Mass Function in CLASS

The RDCC-modified halo mass function is computed as

$$\frac{dn_{\text{RDCC}}}{dM} = \frac{dn_{\Lambda\text{CDM}}}{dM} \left[1 - \chi_{\text{HMF}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (50)$$

CLASS must:

- compute the standard Sheth–Tormen HMF,
- apply the RDCC suppression factor,
- tabulate dn/dM on a logarithmic mass grid,
- store the result in `background->rdcc_egf`.

B.5 Cooling Thresholds and Gas Temperatures

The molecular cooling threshold becomes

$$M_{\text{cool,RDCC}} = M_{\text{cool}} \left[1 + \chi_{\text{cool}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (51)$$

CLASS must:

- compute $T_{\text{vir}}(M, z)$,
- compute H_2 cooling rates using RDCC-modified densities,
- evaluate the RDCC cooling threshold,
- store the result in `thermo->rdcc_cool`.

B.6 Pop III and Pop II Star Formation Rates

The Pop III SFR is

$$\dot{\rho}_{\star,\text{III,RDCC}} = \dot{\rho}_{\star,\text{III}} \left[1 - \chi_{\text{III}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (52)$$

The Pop II SFR is

$$\dot{\rho}_{\star,\text{II,RDCC}} = \dot{\rho}_{\star,\text{II}} \left[1 - \chi_{\text{II}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (53)$$

CLASS must:

- compute SFRs on the RDCC-modified HMF grid,
- integrate over halo mass,
- store the results in `background->rdcc_sfr`.

B.7 Stellar Mass and UV Luminosity

The stellar mass becomes

$$M_{\star,\text{RDCC}} = M_{\star} \left[1 - \chi_{\star} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (54)$$

The UV luminosity function becomes

$$\phi_{\text{RDCC}}(L) = \phi(L) \left[1 - \chi_{\text{UV}} \left(\frac{L}{L_{\text{rel}}} \right)^{-\alpha} \right]. \quad (55)$$

CLASS must:

- compute $M_{\star}(M, z)$,
- convert SFR to UV luminosity,
- tabulate the UV-LF on a luminosity grid,
- output the UV-LF via `output.c`.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic mass grids for HMF and SFR,
- spline-interpolate all RDCC-EGF quantities,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small masses,
- ensure smooth transitions between Pop III and Pop II regimes.

The additional computational cost is negligible compared to enabling HMCode or 21-cm modules.

B.9 Summary

This appendix provides a complete implementation strategy for the RDCC early-galaxy formation model in CLASS. The modifications are minimal, the numerical cost is low, and the resulting stellar masses, star formation rates, and UV luminosity functions are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Comparison with JWST and High- z Galaxy Observations

This appendix compares the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) for early galaxy formation with current observational constraints from JWST, HST legacy fields, quasar damping wings, and indirect 21-cm limits. These observations probe the stellar mass buildup, UV luminosity function (UV-LF), star formation rates (SFRs), clustering, and metallicity of galaxies at $z > 6$.

C.1 UV Luminosity Function

JWST observations show:

- a suppressed faint-end slope at $z > 10$,
- a slower evolution of the UV-LF than predicted by Λ CDM,
- fewer low-luminosity galaxies than expected.

RDCC predicts:

- reduced star formation efficiency in low-mass halos,
- delayed cooling and gas accretion,
- suppressed faint-end UV-LF due to the RDCC HMF suppression.

Key agreement. RDCC naturally reproduces the shallow faint-end slope and slow evolution of the UV-LF observed by JWST without requiring additional feedback or tuning.

C.2 Stellar Masses and Star Formation Rates

JWST reveals:

- lower stellar masses at $z > 10$ than predicted by Λ CDM,
- reduced specific star formation rates (sSFRs),
- smoother stellar mass buildup.

RDCC predicts:

- reduced baryon fractions in low-mass halos (Companion XX),
- delayed Pop III and Pop II star formation (Section 4),
- slower stellar mass growth (Section 5).

Key agreement. RDCC matches the suppressed stellar masses and reduced sSFRs observed by JWST.

C.3 Galaxy Clustering

JWST and HST clustering analyses show:

- enhanced clustering of galaxies at $z > 8$,
- bias values higher than expected for their stellar masses,
- strong large-scale correlations.

RDCC predicts:

- enhanced galaxy bias due to suppressed $\sigma(M)$ (Section 5),
- stronger clustering for halos near the relaxation scale k_{rel} ,
- reduced abundance of low-mass halos, increasing the relative weight of massive halos.

Key agreement. RDCC naturally explains the enhanced clustering of early galaxies without invoking exotic feedback or modified initial conditions.

C.4 Quasar Damping Wings and Metallicity

High-redshift quasars provide constraints on:

- the neutral fraction of the IGM,
- the metallicity of early galaxies,
- the distribution of ionized bubbles.

Observations show:

- moderate metallicities at $z \sim 7$,
- large ionized regions around quasars,
- significant sightline variation.

RDCC predicts:

- delayed but extended metal enrichment (Section 4),
- larger ionized regions due to enhanced escape fractions (Companion XXI),
- smoother damping wings due to reduced small-scale structure.

Key agreement. RDCC matches the observed metallicities and ionized bubble sizes without requiring extreme Pop III star formation.

C.5 Indirect 21-cm Constraints

Although no detection exists yet, 21-cm upper limits imply:

- no strong early heating,
- no deep absorption trough at $z \sim 15\text{--}20$,
- delayed coupling of the spin temperature.

RDCC predicts:

- reduced early star formation (Section 4),
- delayed Pop III heating,
- weaker global 21-cm absorption (Companion XXI).

Key agreement. RDCC is consistent with all current 21-cm upper limits.

C.6 Summary Table

Observable	Data	RDCC Prediction	Agreement
UV-LF faint-end slope	shallow	suppressed faint-end	strong
Stellar masses	low at $z > 10$	reduced buildup	strong
sSFR	low	reduced SFE	strong
Clustering	enhanced	higher bias	strong
Metallicity	moderate	delayed enrichment	strong
21-cm limits	weak heating	delayed Pop III	strong

Table 1: Comparison of RDCC predictions with JWST and high- z galaxy observations.

C.7 Concluding Remarks

RDCC differs fundamentally from Λ CDM in that:

- early galaxies form later and more smoothly,
- stellar masses and SFRs are reduced at $z > 10$,
- the UV-LF faint-end slope is suppressed,
- early galaxies are more strongly clustered,
- metal enrichment is delayed,
- and the relaxation scale k_{rel} imprints a characteristic signature.

These differences provide clear observational signatures for current and upcoming surveys, especially JWST, Roman, and 21-cm experiments.

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